



CSEE4119: Computer Networks

Midterm Preview

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Slides credited to J.F Kurose and K.W. Ross and their textbook "Computer Networking: A Top-Down Approach"

Recap

■ Basics

- Protocol
- Circuit switching vs packet switching
- Performance metrics: transmission rate/time, packet delay, throughput, packet loss rate
- Layers: modular functionality, headers

Recap

- Application layer
 - Client-server, peer-to-peer
 - Sockets
 - Web: persistent, non-persistent HTTP connection; web cache
 - DNS
 - Video streaming
 - P2P: BitTorrent, file distribution time analysis, DHT

Recap

- Transport layer
 - UDP: connection-less, minimalist
 - TCP: Reliable data transfer
 - Mechanisms to deal with bit errors, packet delay, packet losses, out-of-order arrivals, duplicated packets
 - GBN vs selective repeat
 - TCP: Congestion control
 - End-host based vs network-assisted congestion control
 - TCP main mechanisms: slow start, AIMD
 - Packet-loss based vs packet-delay based
 - Mainstream: CUBIC, BBR

Example Question

#1: Consider host A trying to send a 14.6MB photo to host B using TCP. If the TCP header is 20 bytes, IP header is 20 bytes, how many packets would host A needs to send out to transmit the whole photo? What is the overall packet header overhead? (The maximal IP packet is 1500 bytes)

A: The maximal data each IP packet can contain is $(1500 - 20 - 20) = 1460$ bytes. Hence the photo will be segmented into $14.6\text{MB}/1.46\text{KB} = 10^4$ packets.

Total header overhead is $10^4 \times (20+20) = 400 \text{ KB}$.

Example Question

#2: Your host are sending a message of 10 packets of 0.012Mb each to a server, and the server immediately processes your request once it arrives in full. It replies with negligible latency by sending 1 packet of 0.012 Mb.

Your network card connects to a router, which then connects to the server. Both links have 6 Mbps bandwidth, with a negligible latency or propagation delay. The router uses store-and-forward transmission. What is the time between you sending out the first bit and receiving the last?

A: Each packet needs $0.012/6 = 2\text{ms}$ to be transmitted over a link. Because the router uses store-and-forward transmission, after you finish sending 10 packets, 9 of them already arrived at the server. the 10th packet will need additional 2ms to travel from the router to the server. After receiving all packets, server's reply (0.012Mb) will need 2ms to travel to the router, and another 2ms to reach your computer. Thus, the total time span is $20 + 2 + 2 + 2 = 26\text{ms}$.

Example Question

#3: Calculate the # of RRT needed to fetch and receive an index.html with 5 embedded jpeg images. Assume each image can fit in one packet.

- (1) HTTP 1.0 with no parallel TCP connection.
- (2) HTTP 1.1 w/ persistent connections and no pipelining.
- (3) HTTP 1.1 w/ persistent connections and pipelining.

A: (1) $6 * 2RTT = 12 RTT$

(2) $2 + 5 * 1 = 7 RTT$

(3) $2 + 1 = 3 RTT$

Example Question

#4: We learned TCP's estimation of RTT using EWMA. Sender updates its estimated RTT based on each segment transmission `SampleRTT`. TCP avoids measuring `SampleRTT` for retransmitted segments. What do you think is the rationale behind this?

A: If the original segment did arrive and it just took longer, then the ACK arriving next will be an ACK for the original segment transmission. The sender will mistake this as the ACK for retransmission and hence mess up the running average of RTT.

Example Question

#5: Suppose you'd like to register a new domain named columbia.fun.edu, and you wish to set up grasshopper.cs.columbia.edu (129.170.214.115) as a server. What DNS records should you register?

A: Register NS and A records with the authority for the .edu TLD:
(columbia.fun.edu, grasshopper.cs.columbia.edu, NS)
(grasshopper.cs.columbia.edu, 129.170.214.115, A)